

Skills required

- Knowledge of typical DSP theoretical aspects like sampling, convolution, correlation, filters, forward error-correction codes, speech codecs, digital modulation techniques and FFT
- DSP architectures (single-core and multi-core)
- C/Assembly programming on real-time OS
- Fixed-point representation of numbers
- Hardware accelerators
- Hands-on experience with tools like MATLAB
- Exposure to equipment like spectrum analysers, signal generators and modulators

—Sunil Kshatri, AVP, human resources, Sasken

Common resume keywords for DSP jobs

Algorithms, architecture, C, C++, communications, digital, DSP, embedded, hardware, MATLAB, maths, real-time, signal processing, simulation, wireless, ASIC, FPGA, image processing, RTOS

understanding of different platforms and an attitude to learn are the minimum requirements. He says, "The candidate must have hunger and passion to learn and face challenges in this particular domain."

What's in it for freshers?

"Some of the more advanced techniques are not taught in courses. So a fresher should view the first two-three years of his career as an apprenticeship in the area," says Gandhe. Once the engineer achieves a level of expertise, career growth is very fast.

"For a fresher who has moved into signal processor domain, opportunities mostly start from implementing an algorithm for exactness and later on optimising it for memory, MIPS, performance, power, etc," informs Srinivas.

He says, "As the career progresses, engineers should explore opportunities to develop the algorithm for specific applications—be it noise cancellation for a specific environment or a new communication algorithm for a proprietary communication network."

"This provides good challenges and opportunities to showcase the experience an engineer gained over a period of time to develop new IP and demonstrate his skills," he adds.

There are opportunities for freshers but they must fulfil the basic requirement to enter DSP domain. Kaushal explains, "Freshers must have good analytical skills. The assigned task depends on the capabilities of the candidate at entry level. There are in-house training programmes organised by companies to train candidates in DSP field as per the working structure of the company."

George informs that freshers are offered profiles in the areas of software development, firmware debugging, audio codec, RTL implementation of DSP algorithms, and system design and verification. The operative tasks that they would be performing are design, development, documentation, review, integration and testing of physical-layer signal processing and control modules on DSP/RISC processors.

Pay package

The pay package depends upon the skills and expertise of the candidate. "In current market, the average pay package ranges from ₹ 250,000 to ₹ 500,000, depending on the capabilities of the candidate," informs Kaushal.

Training

Talking about the training programme at Analog Devices India, Giri says, "We have a reputation of going to the top campuses every year and hiring good talent via rigorous written and interview process. On joining Analog Devices, candidates go through a rigorous programme where they use Analog Devices' DSP processors and program them to gain the necessary knowledge. They implement key al-

gorithms and gain mastery hands-on. On joining a project, they continue to enhance their skill set."

Adding to that Srinivas says, "In many institutions, software optimisation methodologies are taught only at the postgraduate level. If a fresher has not studied software optimisation methodologies, we provide some on-the-job training in this area."

Talking about the training atmosphere at Cadence Design Systems, Shukla shares, "We provide on-the-job training on real-life projects. This is the best training that freshers can get, as they are exposed to real designs and real-world pressures. In addition, we regularly have knowledge sharing sessions where experienced designers discuss projects with the team. We also assign mentors to freshers to guide them while they are new to the job."

Companies like Broadcom and Sasken provide a basic overview of DSPs, followed by on-the-job training.

Sharing how eInfochips goes about training freshers in the DSP domain, Kaushal informs, "We have eITRA (eInfochips Training and Research Academy) at eInfochips, which is an initiative to narrow the gap between the academia and the industry. It provides special training programmes in the DSP domain. Also, we have project-specific training programmes in groups, which form the second level of the training process."

To sum up

"DSP domain is growing very fast in India and has a number of lucrative positions and work profiles to offer," says George. It is challenging, but has a lot of potential to grow. Giri says, "My strong recommendation would be to get a master's degree in signal processing as it helps one to build a career. Various institutes offer a number of short-term courses. One needs to understand their offerings and make a wise call on the effectiveness of the courses." ●

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